UNIT –II:

Need of Preprocessing, Data cleaning, Data, Data Integration-Entity Identification Problem, Redundancy and Correlation Analysis, Tuple Duplication, Data Value Conflict Detection and Resolution Data transformation, Data Reduction-Transforms, Principal Components Analysis, Attribute Subset Selection, Regression and Log-Linear Models: Parametric Data Reduction, Data Transformation and Data Discretization -Discretization by Binning, Discretization by Histogram Analysis, Discretization by Cluster and Correlation Analyses, Concept Hierarchy Generation for Nominal Data

Data preprocessing

Data preprocessing describes any type of processing performed on raw data to prepare it for another processing procedure. Commonly used as a preliminary data mining practice, data preprocessing transforms the data into a format that will be more easily and effectively processed for the purpose of the user.

Need of Preprocessing

- Data in the real world can be in **incomplete**, **noisy** and **inconsistent** from. These data needs to be preprocessed in order to help improve the quality of the data, and quality of the mining results.
  - → If no quality data, then no quality mining results. The quality decision is always based on the quality data.
  - → If there is much irrelevant and redundant information present or noisy and unreliable data, then knowledge discovery during the training phase is more difficult

Incomplete data:

- Lacking attribute values, lacking certain attributes of interest, or containing only aggregate data. e.g., occupation=" ".
- Incomplete data may come from
  - "Not applicable" data value when collected
  - Different considerations between the time when the data was collected and when it is analyzed.
  - Human/hardware/software problems

Noisy data:

- Containing errors or outliers data. e.g., Salary="-10"
- Noisy data (incorrect values) may come from
  - Faulty data collection by instruments
  - Human or computer error at data entry
  - Errors in data transmission

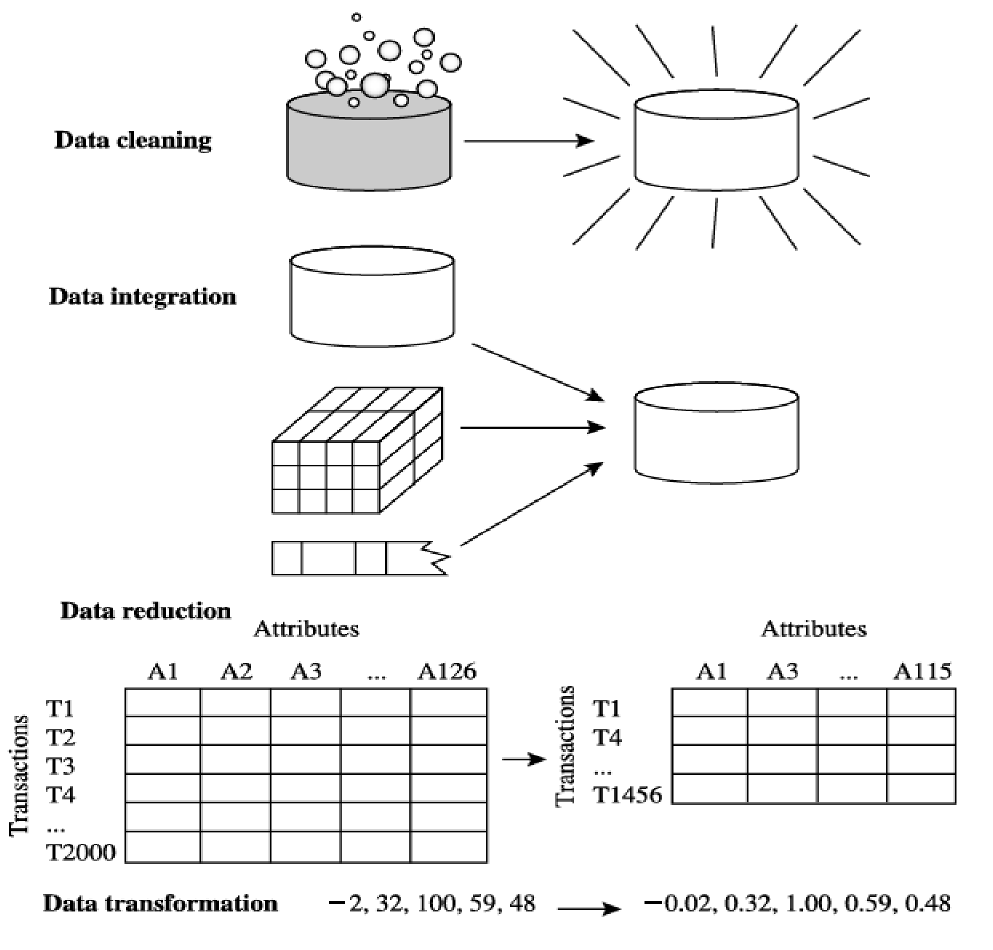
Inconsistent data:

- Containing discrepancies in codes or names.
  - e.g., Age="42" Birthday="03/07/1997"
- Inconsistent data may come from
  - Different data sources
  - Functional dependency violation (e.g., modify some linked data)

Major Tasks in Data Preprocessing

1. ***Data cleaning***
   - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
2. ***Data integration***
   - Integration of multiple databases, data cubes, or files
3. ***Data transformation***
   - Normalization and aggregation
4. ***Data reduction***
   - Obtains reduced representation in volume but produces the same or similar analytical results
5. ***Data discretization***
   - Part of data reduction but with particular importance, especially for numerical data

<u>Forms of Data Preprocessing</u>



Forms of data preprocessing.

Data Cleaning

Data cleaning routines attempt to fill in missing values, smooth out noise while identifying outliers, and correct inconsistencies in the data.

Various methods for handling this problem:

1 Missing Values

The various methods for handling the problem of missing values in data tuples include:

- **Ignoring the tuple**: This is usually done when the class label is missing (assuming the mining task involves classification or description). This method is not very effective unless the tuple contains several attributes with missing values. It is especially poor when the percentage of missing values per attribute varies considerably.

- **Manually filling in the missing value:** In general, this approach is time-consuming and may not be a reasonable task for large data sets with many missing values, especially when the value to be filled in is not easily determined.

- **Using a global constant to fill in the missing value:** Replace all missing **attribute** values by the same constant, such as a label like "Unknown," or $-\infty$. If missing values are replaced by, say, "Unknown," then the mining program may mistakenly think that they form an interesting concept, since they all have a value in common — that of "Unknown." Hence, although this method is simple, it is not recommended.

- **Using the attribute mean for quantitative (numeric) values or attribute mode for categorical (nominal) values, for all samples belonging to the same class as the given tuple:** For example, if classifying customers according to credit risk, replace the missing value with the average income value for customers in the same credit risk category as that of the given tuple.

- **Using the most probable value to fill in the missing value**: This may be **determined** with regression, inference-based tools using Bayesian formalism, or decision tree induction. For example, using the other customer attributes in your data set, you may construct a decision tree to predict the missing values for income.

2 Noisy data:

Noise is a random error or variance in a measured variable. Data smoothing tech is used for removing such noisy data.

<u>Several Data **smoothing** techniques:</u>

1 Binning methods: Binning methods smooth a sorted data value by consulting the neighborhood", or values around it. The sorted values are distributed into a number of 'buckets', or bins. Because binning methods consult the neighborhood of values, they perform local smoothing.

In this technique,

- The data for first sorted
- Then the sorted list partitioned into equi-depth of bins.
- Then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.

Smoothing by bin means: Each value in the bin is replaced by the mean value of the bin.

Smoothing by bin medians: Each value in the bin is replaced by the bin median.

Smoothing by boundaries: The min and max values of a bin are identified as the bin boundaries. Each bin value is replaced by the closest boundary value.

- ➢ Example: Binning Methods for Data Smoothing
  - o Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34
  - o Partition into (equi-depth) bins(equi depth of 3 since each bin contains three values):
    - **Bin 1**: 4, 8, 9, 15
    - **Bin 2**:21, 21, 24, 25
    - **Bin 3**:26, 28, 29, 34
  - o Smoothing by bin means:
    - **Bin 1**: 9, 9, 9, 9
    - **Bin 2**:23, 23, 23, 23
    - **Bin 3**:29, 29, 29, 29
  - o Smoothing by bin boundaries:
    - **Bin 1**: 4, 4, 4, 15
    - **Bin 2**:21, 21, 25, 25
    - **Bin 3**:26, 26, 26, 34

In smoothing by bin means, each value in a bin is replaced by the mean value of the bin. For example, the mean of the values 4, 8, and 15 in Bin 1 is 9. Therefore, each original value in this bin is replaced by the value 9. Similarly, smoothing by bin medians can be employed, in which each bin value is replaced by the bin median. In smoothing by bin boundaries, the minimum and maximum values in a given bin are identified as the bin boundaries. Each bin value is then replaced by the closest boundary value.

Suppose that the data for analysis include the attribute age. The age values for the data tuples are (in increasing order): 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.Use smoothing by bin means to smooth the above data, using a bin depth of 3. Illustrate your steps. Comment on the effect of this technique for the given data. The following steps are required to smooth the above data using smoothing by bin means with a bin depth of 3.

Step 1: Sort the data. (This step is not required here as the data are already sorted.)
Step 2: Partition the data into equi-depth bins of depth

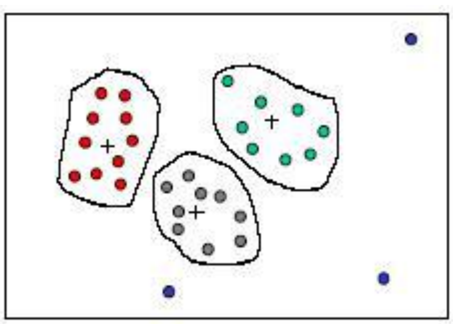
Bin 1: 13, 15, 16
Bin 2: 16, 19, 20
Bin 3: 20, 21, 22
Bin 4: 22, 25, 25
Bin 5: 25, 25, 30
Bin 6: 33, 33, 35
Bin 7: 35, 35, 35
Bin 8: 36, 40, 45
Bin 9: 46, 52, 70

Step 3: Calculate the arithmetic mean of each bin.

Step 4: Replace each of the values in each bin by the arithmetic mean calculated for the bin.

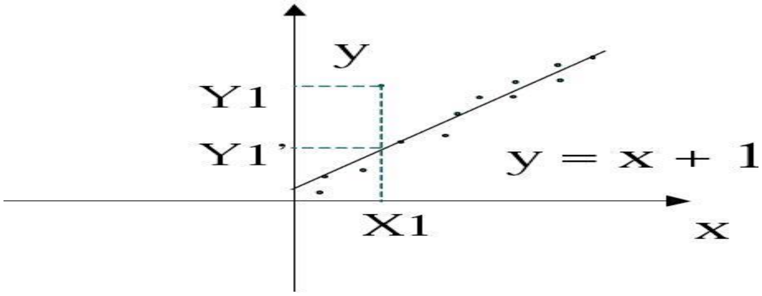
Bin 1: 14, 14, 14
Bin 2: 18, 18, 18
Bin 3: 21, 21, 21
Bin 4: 24, 24, 24
Bin 5: 26, 26, 26
Bin 6: 33, 33, 33
Bin 7: 35, 35, 35
Bin 8: 40, 40, 40
Bin 9: 56, 56, 56

2 Clustering: Outliers in the data may be detected by clustering, where similar **values** are organized into groups, or ‘clusters’. Values that fall outside of the set of clusters may be considered outliers.



3 Regression : smooth by fitting the data into regression functions.

- Linear regression involves finding the best of line to fit two variables, so that



- Multiple linear regression is an extension of linear regression, where more than two variables are involved and the data are fit to a multidimensional surface.
- Using regression to find a mathematical equation to fit the data helps smooth out the noise.

DATA CLEANING AS A PROCESS

- **Field overloading:** is a kind of source of errors that typically occurs when **developers** compress new attribute definitions into unused portions of already defined attributes.
- **Unique rule** is a rule says that each value of the given attribute must be **different** from all other values of that attribute.
- **Consecutive rule** is a rule says that there can be no missing values between the **lowest** and highest values of the attribute and that all values must also be unique.
- **Null rule** specifies the use of blanks, question marks, special characters or **other** strings that may indicate the null condition and how such values should be handled.

Data Integration

It combines data from multiple sources into a coherent store. There are number of issues to consider during data integration.

Issues:

- **Schema integration:**
  - Refers integration of metadata from different sources.
- **Entity identification problem:**
  - Identifying entity in one data source similar to entity in another table.
  - For example, **customer_id** in one database and **customer_no** in another database refer to the same entity

- **Detecting and resolving data value conflicts:**
    - Attribute values from different sources can be different due to different representations, different scales.
        - E.g. metric vs. British units
- **Tuple duplication**
    - **Tuple duplication** refers to the **occurrence of the same record (tuple)** more than once in the integrated dataset. This can happen when **data is combined from multiple sources** and **redundant entries** (i.e., identical or nearly identical rows) are **not identified or removed**.
    -
- **Redundancy:**
    - This is another issue while performing data integration. Redundancy can occur due to the following reasons:
        - **Object identification**: The same attribute may have different names in different database.
        - **Derived Data**: one attribute may be derived from another attribute.

Handling redundant data in data integration

1. Correlation analysis

<u>For numeric data</u>

Some redundancy can be identified by correlation analysis. The correlation between two variables A and B can be measured by

$$r_{A,B} = \frac{\Sigma(A-\overline{A})(B-\overline{B})}{(n-1)\sigma_A \sigma_B}$$

$\overline{A}$, $\overline{B}$ are respective mean values of A and B

σ_A , σ_B are respective standard deviation of A and B

n is the number of tubles

- The result of the equation is > 0, then A and B are positively correlated, which means the value of A increases as the values of B increases. The higher value may indicate redundancy that may be removed.
- The result of the equation is = 0, then A and B are independent and there is no correlation between them.
- If the resulting value is < 0, then A and B are negatively correlated where the values of one attribute increase as the value of one attribute decrease which means each attribute may discourages each other.

For categorical data

- X^2 (chi-square) test

$$\chi^2 = \sum \frac{(Observed - Expected)^2}{Expected}$$

- The larger the X^2 value, the more likely the variables are related
- The cells that contribute the most to the X^2 value are those whose actual count is very different from the expected count
- Correlation does not imply causality
  - # of hospitals and # of car-theft in a city are correlated
  - Both are causally linked to the third variable: population

Example:

	Play chess	Not play chess	Sum (row)
Like science fiction	250(90)	200(360)	450
Not like science fiction	50(210)	1000(840)	1050
Sum(col.)	300	1200	1500

- X^2 (chi-square) calculation (numbers in parenthesis are expected counts calculated based on the data distribution in the two categories)

$$\chi^2 = \frac{(250-90)^2}{90} + \frac{(50-210)^2}{210} + \frac{(200-360)^2}{360} + \frac{(1000-840)^2}{840} = 507.93$$

- It shows that like_science_fiction and play_chess are correlated in the group

Table Breakdown:

	Male	**Female**	**Total**
Fiction	250	200	450
Non-Fiction	50	1000	1050
Total	300	1200	1500

The numbers in parentheses (e.g., 250 (90)) are **expected counts** under the assumption that **gender and book preference are independent**. We'll focus on observed values only for degrees of freedom.

Degrees of Freedom:

This is a **2×2 contingency table**, so:

$$df=(R-1)(C-1)=(2-1)(2-1)=1$$

So **only one cell is free to vary** — all others are determined by **row and column totals**.

Which Cell Can Vary Freely?

Let's pick **the top-left cell: fiction & male = 250**.

Given the row and column totals, here's how **all other cells are determined**:

1. **Row total (Fiction)** = 450 →
   Fiction & Female = 450 − 250 = **200**
2. **Column total (Male)** = 300 →
   Non-Fiction & Male = 300 − 250 = **50**
3. **Row total (Non-Fiction)** = 1050 →
   Non-Fiction & Female = 1050 − 50 = **1000**

So, by setting **just one value** (say 250), the remaining three are **automatically fixed**.

Data Transformation

Strategies of Data Transformation

- In *data transformation*, the data are transformed or consolidated into forms appropriate for mining.
- Strategies for data transformation include the following:
  1. **Smoothing**, works to remove noise from the data. Techniques include ***binning***, ***regression***, and ***clustering***.
  2. **Attribute construction** (or *feature construction*), where new attributes are constructed and added from the given set of attributes to help the mining process.
  3. **Aggregation**, where summary or aggregation operations are applied to the data.
     - For example, the daily sales data may be aggregated so as to compute monthly and annual total amounts. This step is typically used in constructing a data cube for data analysis at multiple abstraction levels.
  4. **Normalization**, where the attribute data are scaled so as to fall within a smaller range, such as -1.0 to 1.0, or 0.0 to 1.0.
  5. **Discretization**, where the raw values of a numeric attribute (e.g., *age*) are replaced by interval labels (e.g., 0–10, 11–20, etc.) or conceptual labels (e.g., *youth*, *adult*, *senior*).
  6. **Concept hierarchy generation for nominal data**, where attributes such as *street* can be generalized to higher-level concepts, like *city* or *country*. Many hierarchies for nominal attributes are implicit within the database schema and can be automatically defined at the schema definition level.

Data Transformation by Normalization

- → The measurement unit used can affect the data analysis.
  - For example,
    - Changing measurement units from **meters** to **inches** for ***height***, or from **kilograms** to **pounds** for ***weight***, may lead to very different results.
    - In general, expressing an attribute in smaller units will lead to a larger range for that attribute, and thus tend to give such an attribute greater effect or "weight."
- → To help avoid dependence on the choice of measurement units, the data should be *normalized* or *standardized*.
- → This involves transforming the data to fall within a smaller or common range such as [-1, 1] or [0.0, 1.0]
- → Normalizing the data attempts to give all attributes an equal weight.
- → Normalization is particularly useful for **classification** algorithms involving neural networks or distance measurements such as nearest-neighbor classification and **clustering**.
- → If using the neural network back propagation algorithm for classification mining ***normalizing the input values*** for each attribute measured in the training tuples will help ***speed up the learning phase***.
- → For ***distance-based methods***, normalization helps prevent attributes with initially large ranges (e.g., *income*) from outweighing attributes with initially smaller ranges (e.g., binary attributes). It is also useful when given no prior knowledge of the data.
- → There are many methods for data normalization.
  - *min-max normalization,*
  - z-*score normalization,* and
  - *normalization by decimal scaling*

Min-max normalization

- → Performs a **linear transformation** on the original data.
- → Suppose that *minA* and *maxA* are the minimum and maximum values of an attribute, *A*. Min-max normalization maps a value, v_i , of *A* to v'_i in the range [*new minA*,*new maxA*] by computing

$$v'_i = \frac{v_i - min_A}{max_A - min_A}(new\_max_A - new\_min_A) + new\_min_A.$$

- → Min-max normalization ***preserves the relationships*** among the original data values.
- → It will encounter an "***out-of-bounds***" error if a future input case for normalization falls outside of the original data range for *A*.

Example 3.4 **Min-max normalization.** Suppose that the minimum and maximum values for the attribute *income* are \$12,000 and \$98,000, respectively. We would like to map *income* to the range [0.0, 1.0]. By min-max normalization, a value of \$73,600 for *income* is transformed to $\frac{73{,}600 - 12{,}000}{98{,}000 - 12{,}000}(1.0 - 0) + 0 = 0.716$. ■

z-score normalization (or *zero-mean normalization*),

- → The values for an attribute *A*, are normalized based on the
  - o mean (i.e., average) and
  - o standard deviation of *A*.
- → A value, *vi* , of *A* is normalized to v'_i by computing

$$v'_i = \frac{v_i - \bar{A}}{\sigma_A},$$

where $\bar{A}$ and σ_A are the mean and standard deviation, respectively, of attribute *A*. The mean and standard deviation were discussed in Section 2.2, where $\bar{A} = \frac{1}{n}(v_1 + v_2 + \cdots + v_n)$ and σ_A is computed as the square root of the variance of *A* (see Eq. (2.6)). This method of normalization is useful when the actual minimum and maximum of attribute *A* are unknown, or when there are outliers that dominate the min-max normalization.

Example 3.5 **z-score normalization.** Suppose that the mean and standard deviation of the values for the attribute *income* are \$54,000 and \$16,000, respectively. With z-score normalization, a value of \$73,600 for *income* is transformed to $\frac{73{,}600 - 54{,}000}{16{,}000} = 1.225$. ■

A variation of this z-score normalization replaces the standard deviation of Eq. (3.9) by the *mean absolute deviation* of *A*. The *mean absolute deviation* of *A*, denoted s_A, is

$$s_A = \frac{1}{n}(|v_1 - \bar{A}| + |v_2 - \bar{A}| + \cdots + |v_n - \bar{A}|). \quad (3.10)$$

Thus, z-score normalization using the mean absolute deviation is

$$v'_i = \frac{v_i - \bar{A}}{s_A}.$$

The mean absolute deviation, s_A, is more robust to outliers than the standard deviation, σ_A. When computing the mean absolute deviation, the deviations from the mean (i.e., $|x_i - \bar{x}|$) are not squared; hence, the effect of outliers is somewhat reduced.

Normalization by decimal scaling normalizes by moving the decimal point of values of attribute A. The number of decimal points moved depends on the maximum absolute value of A. A value, v_i, of A is normalized to v'_i by computing

$$v'_i = \frac{v_i}{10^j},$$

where j is the smallest integer such that $max(|v'_i|) < 1$.

Example 3.6 **Decimal scaling.** Suppose that the recorded values of A range from -986 to 917. The maximum absolute value of A is 986. To normalize by decimal scaling, we therefore divide each value by 1000 (i.e., $j = 3$) so that -986 normalizes to -0.986 and 917 normalizes to 0.917.

Note that normalization can change the original data quite a bit, especially when using z-score normalization or decimal scaling. It is also necessary to save the normalization parameters (e.g., the mean and standard deviation if using z-score normalization) so that future data can be normalized in a uniform manner.

Data Discretization

Data transformation and discretization are essential data pre-processing techniques used in data mining and machine learning. They prepare raw data for analysis by converting it into a more suitable format.

1. Discretization by Binning

- Binning is a ***top-down***, unsupervised data discretization method.
- It divides the range of a continuous attribute into a number of intervals, or "bins."
- Each bin is then assigned a label, often an integer or a categorical value, representing the data points it contains.

Methods of Binning:

- **Equal-Width (Equi-Width) Binning:**
  - The range of the attribute is divided into k equal-sized intervals.
  - The width of each bin is calculated as: w = (max−min)/k, where max and min are the maximum and minimum values of the attribute, and k is the number of bins.
  - **Example:**Consider the age data: 13, 15, 16, 20, 24, 25, 28, 30, 31, 35, 38, 45, 52.
    - Min = 13, Max = 52. Let's use k=3 bins.
    - Width = (52−13)/3=13.
    - **Bin 1:** [13,26]
    - **Bin 2:** [26,39]
    - **Bin 3:** [39,52]

  Diagram:

  ```
  |----- Bin 1 -----|----- Bin 2 -----|----- Bin 3 -----|
  13                26                39              52
  ```
- **Equal-Frequency (Equi-Depth) Binning:**
  - The attribute values are sorted, and the total number of values is divided by the number of bins, k.
  - Each bin contains an approximately equal number of data points.
  - **Example:** Using the same age data: 13, 15, 16, 20, 24, 25, 28, 30, 31, 35, 38, 45, 52 (13 values). Let's use k=3 bins.
    - Each bin should contain approximately 13/3 approx4−5 values.

- **Bin 1:** 13, 15, 16, 20
- **Bin 2:** 24, 25, 28, 30, 31
- **Bin 3:** 35, 38, 45, 52

Diagram:

```
Bin 1   Bin 2       Bin 3
|----|-------|-------------|
13   24      35            52
```

(Each bin contains a similar number of data points)

- **Binning by Clustering:**
  - Data points are grouped into clusters, and each cluster represents a bin.
  - This is a more sophisticated method that considers the distribution of the data.

2. Discretization by Histogram Analysis

- Histograms are a visual representation of the distribution of a numerical attribute.
- Discretization by histogram analysis involves partitioning the data based on the frequency distribution shown in the histogram.

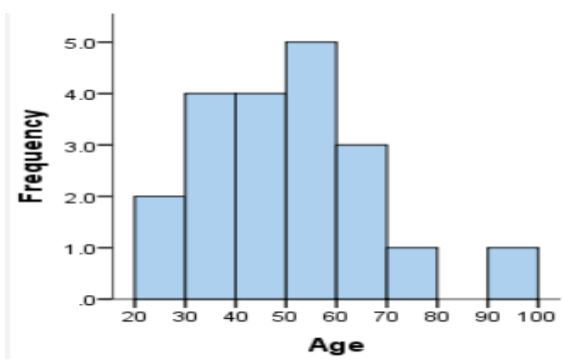
Purpose: To group a large set of values into a smaller set of intervals.

- **Process:**
  1. Create a histogram for the ***continuous*** attribute.
  2. Each bar (bucket) in the histogram can be treated as a bin.
  3. The boundaries of the bins are determined by the histogram's structure.
- **Types of Histograms:**
  - **Equi-width:** Similar to equal-width binning, ***all bins have the same width***.

  - **Equi-depth:** Bins have an approximately ***equal number of data points***.

  - **V-Optimal:** A more advanced method that ***seeks to minimize the variance within each bin.***

Diagram:



- Each bar in the histogram represents a discrete interval.

3. Discretization by Cluster and Correlation Analyses

These are more advanced, bottom-up methods that group similar values together based on their proximity and relationships.

- **Discretization by Cluster Analysis:**
  - A clustering algorithm (e.g., K-Means) is applied to the data.
  - The resulting clusters are treated as the new discrete intervals.
  - **Example:** For a variable income, a clustering algorithm might identify three clusters: "low income," "medium income," and "high income." These clusters become the discrete categories.
  - This method is powerful because it finds natural groupings in the data.
- **Discretization by Correlation Analysis (Chi2):**
  - This is a supervised method that uses the chi-squared statistic to determine the best bin boundaries.

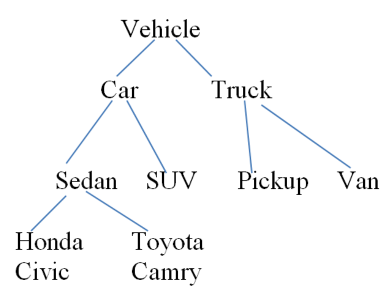
- The goal is to find the discretization that best preserves the relationship between the attribute being discretized and the class labels.
- It merges adjacent intervals that have similar class distributions, thereby maximizing the chi-squared value between the intervals.
- This method is often used for classification tasks.

4. Concept Hierarchy Generation for Nominal Data

A concept hierarchy organizes attributes into a hierarchy of more general and specific concepts. This is particularly useful for nominal (categorical) data.

- **Purpose:** To reduce the number of distinct values for a given nominal attribute, making the data easier to analyze and understand.
- **Methods:**
  1. **Explicitly Defined Hierarchy:** A domain expert defines a hierarchy based on their knowledge.
     - **Example:** For the attribute location, a hierarchy could be: street -> city -> state -> country.
  2. **Schema-Level Hierarchy:** A schema-level hierarchy for a database can be used.
     - **Example:** In a relational database, address could be broken down into street_number, street_name, city, state, and zip_code. The hierarchy is defined by the table structure.
  3. **Using the Number of Distinct Values:** If there are many distinct values, a hierarchy can be generated automatically.
     - **Example:** For an attribute like product_id, the number of distinct values can be used to decide how to group them into higher-level concepts like product_category.
  4. **Automatic Generation:** Some algorithms can automatically generate concept hierarchies by analyzing the data and its distribution.
- **Example of a Concept Hierarchy:**
  - **Level 0 (Specific):** Honda Civic, Toyota Camry, Ford F-150
  - **Level 1 (General):** Sedan, Truck
  - **Level 2 (More General):** Car, Vehicle

Diagram:



<u>**Attribute Subset Selection**</u>

- Attribute Subset Selection is the process of identifying and removing irrelevant, weakly relevant, or redundant attributes from the dataset, helping to improve learning efficiency, reduce overfitting, and enhance model accuracy.

Heuristic Search in Attribute Subset Selection

Because an exhaustive search of all possible attribute subsets is **computationally infeasible** (2^n combinations for n attributes), heuristic methods are used, such as:

1. *Forward Selection*

Stepwise forward selection:

1. *The procedure starts with an empty set of attributes as the reduced set.*
2. *The best of the original attributes is determined and added to the reduced set.*
3. *At each subsequent iteration or step, the best of the remaining original attributes is added to the set.*

2. *Backward Elimination*

Stepwise backward elimination:

1. *The procedure starts with the full set of attributes.*
2. *At each step, it removes the worst attribute remaining in the set.*

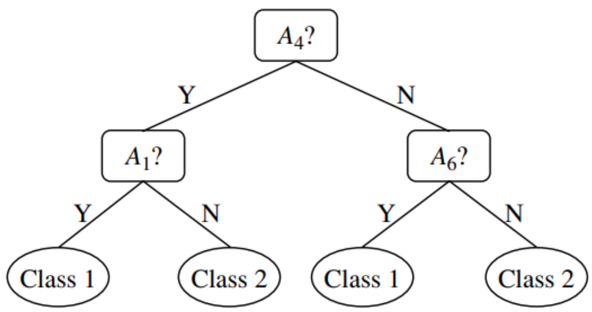
3. Combination of forward selection and backward elimination:

The stepwise forward selection and backward elimination methods:

- ***can be combined so that, at each step, the procedure selects the best attribute and removes the worst from among the remaining attributes.***

4. *Decision Tree Induction*

- Selects attributes during the tree-building process based on split quality (e.g., Information Gain).

Forward selection	Backward elimination	Decision tree induction
Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ Initial reduced set: { } => $\{A_1\}$ => $\{A_1, A_4\}$ => Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ => $\{A_1, A_3, A_4, A_5, A_6\}$ => $\{A_1, A_4, A_5, A_6\}$ => Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$  => Reduced attribute set: $\{A_1, A_4, A_6\}$

Greedy (heuristic) methods for attribute subset selection.

<u>**Regression and Log-Linear Models: Parametric Data Reduction**</u>

- Regression and log-linear models can be used to approximate the given data
- In (simple) linear regression, the data are modeled to fit a straight line.
- For example, a random variable, y (called a response variable), can be modeled as a linear function of another random variable, x (called a predictor variable), with the equation y = wx + b,

Where,.

→ x and y are numeric database attributes.
→ The coefficients, w and b specify the slope of the line and the y-intercept, respectively
→ These coefficients can be solved for by the method of least squares, which minimizes the error between the actual line separating the data and the estimate of the line.

- **Multiple linear regression** is an extension of (simple) linear regression, which allows a response variable, y, to be modelled as a linear function of two or more predictor variables.
- **Log-linear models**
  - Approximate discrete multidimensional probability distributions.
  - Log-linear models can be used to estimate the probability of each point in a multidimensional space for a set of discretized attributes, based on a smaller subset of dimensional combinations.
  - This allows a higher-dimensional data space to be constructed from lower-dimensional spaces.
  - Log-linear models are therefore also useful for **dimensionality reduction** (since the lower-dimensional points together typically occupy less space than the original data points) and **data**

smoothing (since aggregate estimates in the lower-dimensional space are less subject to sampling variations than the estimates in the higher-dimensional space).

- **Regression and log-linear** models can both be used on sparse data, although their application may be limited. While both methods can handle skewed data, regression does exceptionally well.
    - Skewed data refers to a dataset where the data is not evenly distributed, resulting in an asymmetrical distribution with a longer tail on one side. This means the data is not normally distributed and is not symmetrical around its mean.
- Regression can be computationally intensive when **applied to high-dimensional data**, whereas log-linear models show good scalability for up to 10 or so dimensions
- Several software packages exist to solve regression problems.
    - Examples include SAS (www.sas.com), SPSS (www.spss.com), and S-Plus (www.insightful.com).

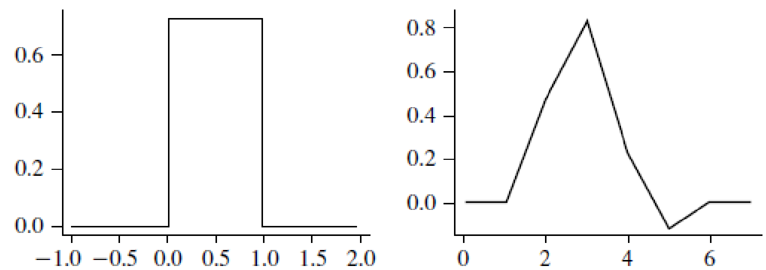
Dimensionality reduction

- is the process of reducing the number of random variables or attributes under consideration.
- which transform or project the original data onto a smaller space
- Dimensionality reduction methods include
    - → *wavelet transforms* and
    - → *principal components analysis*

<u>Wavelet Transforms</u>

- The ***discrete wavelet transform*** **(DWT)** is a linear signal processing technique.
- When it is applied to a data ***vector X***, transforms it to a numerically different ***vector, X'***, of ***wavelet coefficients.***
- The two vectors are of the same length.
- When applying this technique to data reduction, we consider each tuple as an **n-dimensional** data vector, that is, ***X* =(.*x*1,*x*2, : : : ,*xn*),** depicting ***n*** measurements made on the tuple from **n** database attributes.
- We **keep only the strongest wavelet coefficients** (those above a threshold) to compress the data.
- The rest are set to **zero**, resulting in a **sparse** and efficient data representation.
- This helps in **fast computation** and **removes noise** without losing key features.
- The original data can be **reconstructed** using the **inverse of the DWT**
- The DWT is closely related to the ***discrete Fourier transform (DFT),*** a signal processing technique involving ***sines and cosines***.
- In general, however, the DWT achieves better lossy compression.
    - That is, if the same number of coefficients is retained for a DWT and a DFT of a given data vector, the DWT version will provide a more accurate approximation of the original data.
- Hence, for an equivalent approximation, the DWT requires less space than the DFT.
- Unlike the DFT, wavelets are quite localized in space, contributing to the conservation of local detail.
- There is only **one** DFT, yet there are several families of DWTs.
- Popular wavelet transforms include the
    - Haar-2,
    - Daubechies-4, and
    - Daubechies-6.
- The general procedure for applying a discrete wavelet transform uses a **hierarchical *pyramid algorithm*** that halves the data at each iteration, resulting in fast computational speed.
- The method is as follows:
    1. The length, *L*, of the input data vector must be an integer power of 2. This condition can be met by padding the data vector with zeros as necessary ($L \geq n$).

2. Each transform involves applying two functions. The first applies some data smoothing, such as a sum or weighted average. The second performs a weighted difference, which acts to bring out the detailed features of the data.
3. The two functions are applied to pairs of data points in ***X***, that is, to all pairs of measurements **(x_{2i}, x_{2i+1})** This results in two data sets of length $L/2$. In general, these represent a smoothed or low-frequency version of the input data and the high frequency content of it, respectively.
4. The two functions are recursively applied to the data sets obtained in the previous loop, until the resulting data sets obtained are of length 2.
5. Selected values from the data sets obtained in the previous iterations are designated the wavelet coefficients of the transformed data.



(a) Haar-2 (b) Daubechies-4

Figure 3.4 Examples of wavelet families. The number next to a wavelet name is the number of *vanishing moments* of the wavelet. This is a set of mathematical relationships that the coefficients must satisfy and is related to the number of coefficients.

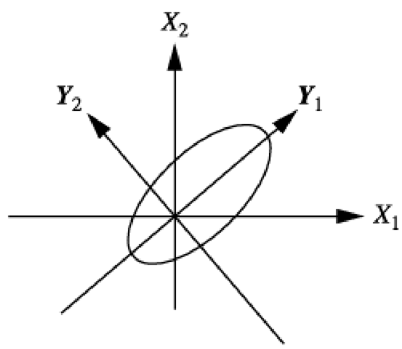
- → Equivalently, a matrix multiplication can be applied to the input data in order to obtain the wavelet coefficients, where the matrix used depends on the given DWT.
- → The matrix must be **orthonormal**, meaning that the columns are unit vectors and are mutually orthogonal, so that the matrix inverse is just its transpose. Although we do not have room to discuss it here, this property allows the reconstruction of the data from the smooth and smooth-difference data sets. By factoring the matrix used into a product of a few sparse matrices, the resulting "fast DWT" algorithm has a complexity of $O(n)$ for an input vector of length n.
- → Wavelet transforms can be applied to multidimensional data such as a data cube.
- → This is done by first applying the transform to the first dimension, then to the second, and so on.
- → The computational complexity involved is linear with respect to the number of cells in the cube.
- → Wavelet transforms give good results on sparse or skewed data and on data with ordered attributes.
- → Lossy compression by wavelets is reportedly better than JPEG compression, the current commercial standard.
- → Wavelet transforms have many real world applications, including the compression of fingerprint images, computer vision, analysis of time-series data, and data cleaning.

Principal Components Analysis

- → Principal components analysis as a method of **dimensionality** reduction.
- → The data to be reduced consist of tuples or data vectors described by n attributes or dimensions.
- → ***Principal components analysis*** (**PCA**; also called the Karhunen-Loeve, or K-L, method) searches for k, n-dimensional orthogonal vectors that can best be used to represent the data, where $k <= n$.
- → The original data are thus projected onto a much smaller space, resulting in dimensionality reduction.
- → PCA "combines" the essence of attributes by creating an alternative, smaller set of variables.
- → The initial data can then be projected onto this smaller set.
- → PCA often reveals relationships that were not previously suspected and there by allows interpretations that would not ordinarily result.

The basic procedure is as follows:

1. The input data are normalized, so that each attribute falls within the same range. This step helps ensure that attributes with large domains will not dominate attributes with smaller domains.
2. PCA computes *k* orthonormal vectors that provide a basis for the normalized input data. These are unit vectors that each point in a direction perpendicular to the others. These vectors are referred to as the *principal components*. The input data are a linear combination of the principal components.
3. The principal components are sorted in order of decreasing "significance" or strength. The principal components essentially serve as a new set of axes for the data, providing important information about variance.
    a. That is, the sorted axes are such that the first axis shows the most variance among the data, the second axis shows the next highest variance, and so on.
    b. For example, Figure 3.5 shows the **first two principal components, *Y*1 and *Y*2,** for the given set of data originally mapped to the axes ***X*1 and *X*2.** This information helps identify groups or patterns within the data.



4. Because the components are sorted in decreasing order of "significance," the data size can be reduced by eliminating the weaker components, that is, those with low variance. Using the strongest principal components, it should be possible to reconstruct a good approximation of the original data.

→ PCA can be applied to ordered and unordered attributes, and can handle sparse data and skewed data. Multidimensional data of more than two dimensions can be handled by reducing the problem to two dimensions.
→ Principal components may be used as inputs to multiple regression and cluster analysis.
→ In comparison with wavelet transforms, PCA tends to be better at handling sparse data, whereas wavelet transforms are more suitable for data of high dimensionality.